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Review Article

OXFORD

SPINE SECTION

Efficacy, Effectiveness, Safety, and Cost-effectiveness of Epidural Adhesiolysis for Treating Failed Back Surgery Syndrome. A Systematic Review

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Abstract

Background. Failed back surgery syndrome (FBSS) has a profound impact on patients' quality of life and represents a major clinical challenge and a significant economic burden for society. Adhesiolysis is used as a treatment to eliminate perineural/epidural adhesions in patients with chronic pain attributed to FBSS. **Objective**. To evaluate the efficacy, effectiveness, safety, and cost-effectiveness of epidural adhesiolysis compared with other procedures for treating FBSS. **Method**. A systematic review was conducted. The electronic databases Medline/PreMedline, EMBASE, Cochrane Library Plus, Centre for Reviews and Dissemination databases, SCOPUS, Science Citation Index, and PEDRO were consulted through April 2017. Predefined criteria were used to determine inclusion of the studies and to assess their methodological quality. **Results**. Ten reports were included. No randomized controlled trials (RCTs) on efficacy or cost-effectiveness were found. Three reports (corresponding to two RCTs, N = 212) suggested that adhesiolysis was effective, especially for pain and disability. However, both studies presented serious methodological flaws. In addition to RCTs, seven observational studies with high risk of bias reported data on effectiveness and safety. Fifty-eight adverse events were reported among 130 patients undergoing endoscopic adhesiolysis, and 19 among the 110 undergoing percutaneous adhesiolysis. **Conclusions**. The evidence on the efficacy and cost-effectiveness of adhesiolysis for treating FBSS is nonexistent, whereas evidence on its effectiveness and safety is insufficient. Incorporating data from observational studies did not improve the quality of the evidence on effectiveness.

Key Words: Systematic Review; Percutaneous Adhesiolysis; Endoscopic Adhesiolysis; Chronic Low Back Pain; Failed Back Surgery Syndrome

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Introduction

Failed back surgery syndrome (FBSS) is defined as persistent radicular and/or lumbar pain after spine surgery. Persistent is usually defined as existing for more than three months, which implies that pain has become chronic [1,2]. FBSS hampers patient quality of life and represents a major clinical challenge and a significant economic burden for society [2–6]. Epidural/peridural fibrosis is a potential cause of FBSS. Excessive fibrosis (i.e., beyond the physiologic scarring process) is believed to tether the dura and the nerve roots, chemically excite and/or compress the nerve roots, damage microvessels, and potentially lead to ischemia, edema, demyelination, and C-fiber activation [3,5,7,8].

Adhesiolysis is the lysis of perineural/epidural adhesions. Percutaneous adhesiolysis was first described as a three-day procedure [9] but was later modified to become a one-day procedure [10]. In routine clinical practice, technical characteristics may vary depending on the practitioner's personal preferences. This includes visualization procedures (e.g., radioscopy, epiduroscopy), products injected for the lysis of adhesions (e.g., hypertonic saline, hyaluronidase, with or without steroids or local anesthetics), and additional interventions performed during the procedure (e.g., pulsed radiofrequency or injections of other substances).

Several reviews claim that adhesiolysis is effective [11–16]. However, few have comprehensively gathered data on safety, not all have complied with the optimal methods [17], and their results disagree on the strength of the available evidence. Moreover, none have focused on FBSS, which may be seen as its major primary indication. Therefore, the purpose of this study is to assess the efficacy, safety, effectiveness, and cost-effectiveness of epidural adhesiolysis for treating patients with chronic pain attributed to FBSS.

Methods

This systematic review followed the recommendations from the PRISMA statement [18], and its protocol was registered in the PROSPERO database. Due to differences in inclusion criteria, in practice, two systematic reviews were conducted at the same time; one on efficacy, effectiveness, and cost-effectiveness, and a second on safety.

Search and Study Selection

The following electronic databases were searched through April 2017: Medline and PreMedline in process, EMBASE, The Cochrane Library Plus (Cochrane Central Register of Controlled Trials, Cochrane Systematic Reviews Database), Centre for Reviews and Dissemination databases (DARE, HTA, NHS-EED), SCOPUS, Science Citation Index, and PEDRO. The search strategy was designed to ensure maximum sensitivity and is shown in Table 1. No date or language restrictions were applied.

Studies were included in this review if they complied with the following criteria:

a. Design: For the review on efficacy, effectiveness, and cost-effectiveness, only randomized controlled trials (RCTs) were eligible. For the review on safety, experimental and observational studies were accepted. If no RCTs were found, or if all of those found turned out to be categorized as "high risk of bias," data from observational studies would be considered as a potential source of data on effectiveness, even if they had no comparator group, provided that they had a sample size $\geq$10 and complied with the other inclusion criteria.
b. Study population: Studies including humans with chronic low back and/or radicular pain caused by FBSS. In the case of studies that included mixed populations (i.e., with FBSS and other diagnosis), only data from patients with FBSS were considered for this review.
c. Intervention: Studies in which at least one group underwent (endoscopic or percutaneous) adhesiolysis.
d. Comparator: For the review on efficacy, effectiveness, and cost-effectiveness, only studies that compared adhesiolysis with other procedures (including control or placebo/sham) were accepted. For the review on safety, both studies with and without a comparator group were accepted.
e. Outcome measures: For the review on efficacy and effectiveness, at least pain or disability had to have been assessed. For the review on safety, any study reporting any type of adverse event was selected. For the review on cost-effectiveness, the study had to report both effectiveness and costs of every comparator and/or the incremental cost-effectiveness ratio.

References that were identified through the electronic search were screened based on their title and abstract by two authors separately, out of a pool of five. The full text of those that were eligible was assessed for inclusion criteria by two authors separately, out of a pool of five. Disagreements at the screening and assessment stages were resolved by consensus with the other authors.

When needed to clarify any aspect of their studies, the corresponding authors from the original studies were contacted, and those who did not respond were followed up with twice within a 14-day interval. References included in the reviewed studies were scanned to identify additional studies.

Data Collection Process, Quality Assessment, and Data Analysis

The methodological quality of the studies included in the review was assessed separately by two reviewers out of a pool of four, and disagreements were solved by consensus with a third reviewer. The methodological quality of RCTs was assessed according to the recommendations from the Cochrane Back Review Group for assessing the risk of bias [17,19], that is, based on whether they complied with a defined set of 11 methodological criteria. A study was categorized as "low risk of bias" when it met six or more of these criteria, although studies with

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Table 1. Search strategies

MEDLINE & PREMEDLINE

#1 exp Low Back Pain/ or exp Failed Back Surgery Syndrome/ or *Pain, Postoperative/
#2 (Post lumbar surgery syndrome or low back pain or epidural adhesion$ or post lumbar laminectomy syndrome or Failed Back Surgery Syndrome).tw.
#3 ("low back" and pain).tw.
#4 (epidural and (fibrosis or scar$)).tw.
#5 Surgery Syndrome.tw.
#6 1 or 2 or 3 or 4 or 5
#7 (hypertonic saline neurolysis or Epidural lysis or 'lysis of adhesions' or Epidural endoscopy or percutaneous adhesiolysis or Racz or spinal endoscopy or Epiduroscopy or epidural adhesion*).tw.
#8 ((Epidural or peridural or lumbar or spinal or low back) and (neuroplasty or neurolysis or epidurolysis or adhesiolysis or lysis)).tw.
#9 ((Non-endoscopic or non endoscopic or endoscopic) and adhesiolysis).tw.
#10 7 or 8 or 9
#11 6 and 10
#12 limit 11 to humans

EMBASE

#1 'low back pain'/exp#
2 'failed back surgery syndrome'/exp
#3 'postoperative pain'/mj
#4 'post lumbar surgery syndrome' OR 'low back pain' OR 'epidural adhesion*' OR 'post lumbar laminectomy syndrome' OR 'failed back surgery syndrome': ab, ti
#5 'low back': ab, ti
#6 pain: ab, ti
#7 #5 AND #6
#8 epidural: ab, ti
#9 fibrosis OR scar OR scars: ab, ti
#10 #8 AND #9
#11 'surgery syndrome': ab, ti
#12 #1 OR #2 OR #3 OR #4 OR #7 OR #10 OR #11
#13 'hypertonic saline neurolysis' OR 'epidural lysis' OR 'lysis of adhesions' OR 'epidural endoscopy' OR 'percutaneous adhesiolysis' OR racz OR 'spinal endoscopy' OR epiduroscopy OR 'epidural adhesion': ab, ti
#14 epidural OR peridural OR lumbar OR spinal OR 'low back': ab, ti
#15 neuroplasty OR neurolysis OR epidurolysis OR adhesiolysis OR lysis: ab, ti
#16 #14 AND #15
#17 'non endoscopic' OR endoscopic: ab, ti
#18 adhesiolysis: ab, ti
#19 #17 AND #18
#20 #13 OR #16 OR #19
#21 #12 AND #20
#22 #12 AND #20 AND [humans]/lim AND [embase]/lim

SCI-EXPANDED/Web of Science

#1 TS=('low back pain' OR 'failed back surgery syndrome' OR 'postoperative pain') OR TI=('post lumbar surgery syndrome' OR 'low back pain' OR 'epidural adhesion$' OR 'post lumbar laminectomy syndrome' OR 'failed back surgery syndrome') OR TI=(("low back" and pain)) OR TI=((epidural and (fibrosis or scar$)))
#2 TI=('hypertonic saline neurolysis' OR 'epidural lysis' OR 'lysis of adhesions' OR 'epidural endoscopy' OR 'percutaneous adhesiolysis' OR racz OR 'spinal endoscopy' OR epiduroscopy OR 'epidural adhesiolysis') OR TI=((Lumbar and (epidurolysis or adhesiolysis))) OR TI=((Epidural and (neuroplasty or neurolysis))) OR TI=(((Non-endoscopic or non endoscopic or endoscopic) and adhesiolysis))
#3 #2 AND #1

(continued)

Table 1. continued

DARE, HTA, NHS-EED

#1 MeSH DESCRIPTOR Low Back Pain EXPLODE ALL TREES
#2 MeSH DESCRIPTOR Failed Back Surgery Syndrome EXPLODE ALL TREES
#3 ('post lumbar surgery syndrome' or 'low back pain' or 'epidural adhesion$' or 'post lumbar laminectomy syndrome' or 'failed back surgery syndrome'): TI
#4 (("low back" and pain)): TI OR ((epidural and (fibrosis or scar$))): TI
#5 #1 OR #2 OR #3 OR #4
#6 (((hypertonic saline neurolysis or Epidural lysis or 'lysis of adhesions' or Epidural endoscopy or percutaneous adhesiolysis or Racz or spinal endoscopy or Epiduroscopy or epidural adhesiolysis)): TI OR ((Lumbar and (epidurolysis or adhesiolysis))): TI OR ((Epidural and (neuroplasty or neurolysis))): TI): TI
#7 ((((Non-endoscopic or non endoscopic or endoscopic) and adhesiolysis))): TI
#8 #6 OR #7
#9 #5 AND #8

COCHRANE

#1 MeSH descriptor: [Low Back Pain] explode all trees
#2 MeSH descriptor: [Failed Back Surgery Syndrome] explode all trees
#3 MeSH descriptor: [Pain, Postoperative] explode all trees
#4 'post lumbar surgery syndrome' or 'low back pain' or 'epidural adhesion$' or 'post lumbar laminectomy syndrome' or 'failed back surgery syndrome': ti, ab, kw (Word variations have been searched)
#5 'low back' and pain: ti, ab, kw or epidural and fibrosis: ti, ab, kw or epidural and scar$: ti, ab, kw (Word variations have been searched)
#6 #1 or #2 or #3 or #4 or #5
#7 'hypertonic saline neurolysis' or 'epidural lysis' or 'lysis of adhesions' or 'epidural endoscopy' or 'percutaneous adhesiolysis' or racz or 'spinal endoscopy' or epiduroscopy or 'epidural adhesiolysis': ti, ab, kw or (Lumbar and (epidurolysis or adhesiolysis)): ti, ab, kw or (Epidural and (neuroplasty or neurolysis)): ti, ab, kw or ((Non-endoscopic or non endoscopic or endoscopic) and adhesiolysis): ti, ab, kw (Word variations have been searched)
#8 #6 and #7

SCOPUS

#1 TITLE-ABS-KEY((Post lumbar surgery syndrome or low back pain or epidural adhesion$ or post lumbar laminectomy syndrome or Failed Back Surgery Syndrome)) AND SUBJAREA(MULT OR MEDI OR NURS OR VETE OR DENT OR HEAL)
#2 TITLE-ABS-KEY(("low back" and pain)) AND SUBJAREA(MULT OR MEDI OR NURS OR VETE OR DENT OR HEAL)
#3 TITLE-ABS-KEY(((epidural AND (fibrosis OR scar$)))) AND SUBJAREA(MULT OR MEDI OR NURS OR VETE OR DENT OR HEAL)
#4 #1 AND #2 AND #3
#5 TITLE-ABS-KEY(((hypertonic saline neurolysis OR epidural lysis OR 'lysis of adhesions' OR epidural endoscopy OR percutaneous adhesiolysis OR racz OR spinal endoscopy OR epiduroscopy OR epidural adhesiolysis)))
#6 TITLE-ABS-KEY (((lumbar AND (epidurolysis OR adhesiolysis))))
#7 TITLE-ABS-KEY (((epidural AND (neuroplasty OR neurolysis))))
#8 TITLE-ABS-KEY (((non-endoscopic OR non endoscopic OR endoscopic) AND (adhesiolysis)))
#9#5 OR #6 OR #7 OR #8
#10 #4 AND #9
#11 #10 AND (LIMIT-TO (DOCTYPE , "ar") OR LIMIT-TO (DOCTYPE , "re")) AND (LIMIT-TO (SRCTYPE , "j"))

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serious flaws were categorized as "high risk of bias" regardless of their score [19]. Were data on effectiveness from observational studies to be included in this review, they would be categorized as "high risk of bias."

The methodological quality of studies on safety was assessed by means of a checklist designed ad hoc based on Centre for Reviews and Dissemination (CRD)'s and EUnetHTA's recommendations [20,21].

Following Cochrane recommendations, data extraction was undertaken separately and in duplicate by two authors out of a pool of four, using standardized electronic forms. For each study, all data on all the variables gathered in each individual study were extracted. The information was summarized through a qualitative synthesis. Two separate meta-analysis on effectiveness were planned; one focusing on pain severity and another on disability. None of the authors involved in quality assessment and data extraction perform adhesiolysis in clinical practice or have any conflicts of interest with regards to this procedure. Results were interpreted by all of the authors, including one expert in performing this procedure in routine clinical practice.

Results

Once duplicates were eliminated, the electronic search produced 513 references, 10 of which were eligible, three reports corresponding to two RCTs [22–24] and seven observational studies [25–31]. Manual search provided no additional references. Figure 1 shows the PRISMA flow chart.

The authors of six studies were contacted to clarify or gather additional information not disclosed in their reports [22–24,32–34]. Only one of them responded, but data made it impossible to ascertain whether all patients included in his study suffered from FBSS, and therefore the study was excluded [32].

Two RCTs on adhesiolysis were excluded because they included patients with and without FBSS, and the leading authors did not respond to requests to provide data restricted to patients with FBSS [33,34].

Efficacy

No RCTs on efficacy of adhesiolysis in patients with chronic pain attributed to FBSS (vs placebo or sham procedures) were found.

Effectiveness

Two single-blind RCTs published in three papers were found [22–24]. They included patients with pain due to FBSS in which conservative treatments had failed for six or more months.

One study was conducted in China [24]. It recruited 92 patients with leg pain (with or without low back pain) and compared percutaneous adhesiolysis (with normal saline and steroids) with an epidural injection of steroids in the anterior epidural space. The study had 17% losses

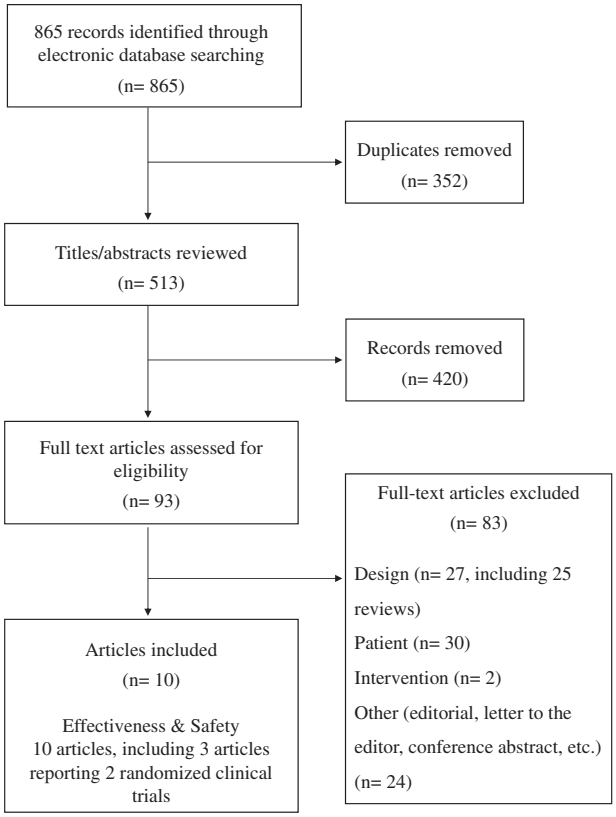

Figure 1. Flow diagram of the selection process of studies.

in both groups after six months of follow-up [24]. The paper did not disclose further details, such as the number of procedures performed in both groups (Table 2).

The other RCT was conducted in the United States [22,23]. It recruited 120 patients with low back pain (with or without leg pain) who were followed up for up to 24 months. All patients continued under drug treatment and exercise programs, as previously prescribed. The RCT compared percutaneous adhesiolysis with epidural injections of normal saline with steroids and anesthetics. Adhesiolysis was repeated after three or more months if the degree of improvement in disability or pain relief experienced after the first procedure deteriorated $\geq$50%. Criteria for repeating epidural injections were not disclosed. Over the follow-up period, the mean (SD) number of procedures was 3.5 (1.0) in the adhesiolysis group and 2.2 (1.1) in the injection group. Losses to follow-up at different time points were up to 10% in the adhesiolysis group and up to 87% in the injection group (Table 2) [23].

Both studies have serious methodological flaws and a high risk of bias. It is unclear whether treatment allocation and outcome assessment were effectively masked (Table 3) [23,24]. In the study by Manchikanti et al., patients could have undergone epidural injections before entering the trial, so it is unclear whether they were actually blinded to the treatment they received. Moreover, patients were allowed to be informed of the treatment they had been assigned to if they asked about it. In this

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Table 2A. Main characteristics of included studies: Randomized clinical trials

First Author Year [Ref]	Country Dates	Methods	Population Inclusion Criteria	Sample No.	Age, Mean (SD), y	Gender, % Females	Pain Duration, Mean (SD), mo	Interventions	No. of Procedures, Mean (SD)	Follow-up	Losses to Follow-up, %
Chun-jing 2012 [24]	China 05/2006–08/2009	Design: single blind ITT analysis Sponsor: NR	>18 y History of lumbar surgery > 6 mo History of chronic function-limiting lower extremity pain ± low back pain of >6 mo (postsurgery) No evidence of facet joint pain Failure to improve substantially with conservative management	46*	59.2 (11.8)	45	13.2 (4.9)	Ventral epidural vascular catheter with 50–80 mL saline and 10 mg dexamethasone	NR	7 d 1 mo 6 mo	6 mo: 17 Other time points: NR
				46*	58.4 (12.3)	40	12.6 (4.8)	Puncture to the epidural anterior space for angiography and an injection of 10 mg dexamethasone	NR	7 d 1 mo 6 mo	6 mo: 17 Other time points: NR
Manchikanti 2012 [23]	USA 01/2006–06/2010	Design: single blind ITT analysis: last available follow-up data or initial data were used in patients with no other data available Sponsor: no external funding	Previous lumbar surgery with or without fusion >6 mo before enrollment History of chronic function-limiting ± low back pain ± extremity pain >6 mo (postsurgery) Failure to improve substantially with conservative management (including transforaminal epidural injections)	60	52 (12.5)	58	196 (109.4)	Percutaneous adhesiolysis with targeted delivery of lidocaine, 10% hypertonic sodium chloride solution, and nonparticulate betamethasone	3.5 (1.0)	3 mo 6 mo 12 mo 18 mo 24 mo	3 mo: 0 6 mo: 1.6 12 mo: 3 18 mo: NR 24 mo: 10
				60	52 (13.9)	58	186 (121.7)	Caudal epidural injections with catheterization up to S3 with local anesthetic, steroids, and 0.9% sodium chloride solution	2.2 (1.1)	3 mo 6 mo 12 mo 18 mo 24 mo	3 mo: 0 6 mo: 16 12 mo: 72 18 mo: NR 24 mo: 87

*It is unclear whether these characteristics refer to the 92 patients who were randomized or to the 72 who were not lost to follow-up. Clarification was requested unsuccessfully from the first author.
EI = epidural injection; ITT = intention-to-treat analysis; NR = not reported; PA = percutaneous adhesiolysis.

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study, the dropout and loss to follow-up rate in the epidural injections group was 70% at 12 months and 87% at 24 months. Moreover, the groups were not similar at baseline in important variables (e.g., opioid use), and analyses did not adjust for these differences [23]. Furthermore, both studies allowed cointerventions (e.g., physiotherapy, acupuncture, exercise, analgesics—including opioids) but did not register them, making it impossible to ascertain whether differences in clinical evolution across groups were actually caused by the interventions or differences in the rate of use of cointerventions [23,24].

Results from both trials were favorable to adhesiolysis. Chun-jing et al. [24] reported high statistically significant differences in terms of pain, disability, opioid use, and the MacNab criteria scale to measure patient satisfaction (Table 4) [35]. Manchikanti et al. [23] reported differences in favor of percutaneous adhesiolysis for pain and disability at three, 18 and 24 months, with no differences in return to work or opioid consumption. Analyses did not adjust for the statistically significant baseline consumption of opioids in the adhesiolysis group. Inconsistencies related to statistical analysis were also found in this report.

The two RCTs used similar methods to assess pain at six months [23,24]. However, it was deemed inappropriate to conduct a meta-analysis because of the high risk of bias the studies presented. For this reason, data on effectiveness derived from observational studies were considered for this review.

In addition to the RCTs, seven observational studies reported data on pain and/or disability [25–31], three of which were retrospective [25,26,31]. As seen in Table 2B, most of these observational studies simply reported the scores for some clinical variables before adhesiolysis was performed, as well as some time afterwards, in small cohorts of patients that were poorly described. In fact, only one (retrospective) study included >60 patients who had actually undergone adhesiolysis [31], and none of the studies included >115. Although all observational studies reported some degree of clinical improvement at some point during the study period, none quantified the use of cointerventions.

Cost-effectiveness

No cost-effectiveness analyses alongside RCTs were found.

Safety

The two RCTs and the seven observational studies reported data on safety for adhesiolysis. In general, methods for gathering data on adverse events were not described in detail, and the methodological quality of these studies was low (Table 3B): sample sizes were small, ranging between 12 and 120; several studies were retrospective [25,26,31]; and only two defined "adverse effects" and reported their seriousness [22,30].

The exact procedure performed varied across studies on safety (e.g., percutaneous vs endoscopic adhesiolysis) (Table 2). Three studies reported no adverse events: one on endoscopic and two on percutaneous adhesiolysis [24,26,29], including one in which destruction of the perineural/epidural adhesions failed in six cases (out of 92 patients) [24]. Another study monitored complications only while the procedure was performed and at the recovery room [31] but did not mention whether they were finally measured or not.

Adverse events reported in the other five studies [22,23,25,27,28,30] included postoperative weakness, sensory deficit, skin rash and itching, significant weight increase, transient hypoacusia, headache and neck pain (attributed to increase of cerebrospinal fluid pressure), wound pain, leg/sciatic pain, dural punction, low back pain, bleeding, and apnea. Infection and subaracnoidal punction and blockade were reported more commonly after endoscopic than after percutaneous adhesiolysis (Table 5) [25]. No data were available with regards to the frequency of these adverse events or the number of patients who reported them. In total, 130 patients underwent endoscopic adhesiolysis in the studies assessing safety, and 58 adverse events were reported. For percutaneous adhesiolysis, 110 patients underwent the procedure and 19 adverse events were reported.

Finally, conflicts of interest were not reported in five studies [25,27–30]. Among the four studies in which they were reported, all of them denied any conflict of interest [23,24,26,31].

Discussion

Patients with FBSS who suffer persistent and refractory pain despite conventional drug treatment face an everyday life characterized by constant pain, disability, and lower quality of life and frustrating expectations of finding relief. Adhesiolysis was designed to help these patients and is based on biologically sound arguments and concepts [36]. To the authors' knowledge, this is the first systematic review to focus on adhesiolysis for treating FBSS specifically.

Unfortunately, there is no evidence on whether this technology is efficacious or cost-effective for this condition, and only two trials with a high risk of bias have explored its effectiveness [22–24].

The seven observational studies identified by this review are also of very low quality and did not provide any data suggesting whether the somewhat positive clinical evolution they depicted could be attributed to adhesiolysis or to any other treatment(s) patients may have been receiving concomitantly, natural history, unspecific effects, or any other reason. Therefore, the quality of the evidence on the potential effectiveness of the procedure

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Table 2B. Main characteristics of included studies: Observational studies

First Author Year [Ref]	Country Dates	Design	Population	Sample No.	Age, Mean (SD), y	Gender, % Female	Pain Duration, Mean (SD), mo	Interventions	No. of Procedures, Mean (SD)	Losses to Follow-up
Manchikanti 1999 [25]	USA	Retrospective	Post–lumbar laminectomy patients who underwent either nonendoscopic adhesiolysis or endoscopic adhesiolysis	60	51.8 (1.76)	37	7 (0.54)	Nonendoscopic adhesiolysis	2.98 + 0.16 (SEM)	NA
				60	48.7 (1.61)	52	8 (0.51)	Endoscopic adhesiolysis	1.28 + 0.07 (SEM)	NA
Abejón 2007 [26]	Spain 05/2005–01/2006	Retrospective	Patients with FBSS with >6 mo of pain and >5 points on a visual pain scale	15	48.3 (7.9)	46	25 (7.9)	Percutaneous adhesiolysis	NR1	NA
Murai 2007 [27]	Japan	Prospective	Patients with low back pain and sciatica with poor response to conservative therapies	37 (FBSS group)	58.0 (15)	35	40 (67)	Endoscopic adhesiolysis	NR1	NR
Avellanal 2008 [28]	Spain	Prospective	Patients (aged 18 y or older) with a history of FBSS with symptoms and signs of severe chronic sciatica (VAS ≥7), defined as pain in the distribution of a lumbar nerve root with or without back pain	19	43.5	57.9	≥1 y	Endoscopic adhesiolysis	NR1	NR
Takeshima 2009 [29]	Japan 06/2001–09/2007	Prospective	FBSS patients in whom symptoms were not sufficiently improved by conservative therapies	28	57.3	17.8	NR	Epiduroscopic adhesiolysis	NR1	NA
Pereira 2016 [30]	Portugal07/2010–10/2012	Prospective	Patients with persistent or recurrent low back and/or lower limb pain (VAS ≥5) after lumbar spine surgery	24	46.6 (9.5)	45.8	≥6 mo	Epiduroscopic adhesiolysis	NR1	NA
Lee 2014 [31]	South Korea 2009–2012	Retrospective	Patients who had undergone PA or transforaminal epidural steroid injection for the treatment of chronic lower back and lower extremity pain ≥3 mo after lumbar surgery	52	56.4 (11.3)	46.1	39.6 (36.3)	Percutaneous adhesiolysis	1	NA
				62	59.2 (13.4)	56.4	36.6 (25.3)	Transforaminal epidural steroid injection	1	NA

EI = epidural injection; FBSS = failed back surgery syndrome; NA = not applicable; NR = not reported; NR1 = not reported but probably only one procedure; PA = percutaneous adhesiolysis; SEM = standard error of the mean; VAS = visual analog scale (range from best to worst = 0–10).

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Table 3A. Methodological quality of the included studies: Risk of bias in the included RCTs, according to recommendations from the Cochrane Back Review Group [19]

		First Author Year [Ref]	
Sources of Risk of Bias		Manchikanti 2012 [23]	Chun-jing 2012 [24]
Randomization	1. Was the method of randomization adequate?	Yes	Yes
Allocation	2. Was treatment allocation masked?	Unclear	No
Blinding patient/provider/outcomes assessor	3. Was the patient blinded to the intervention?	Yes	Yes
	4. Was the care provider blinded to the intervention?	No	No
	5. Was the outcome assessor blinded to the intervention?	Unclear	Unclear
Reporting of the data	6. Was the dropout rate described and acceptable?	No	Yes
	7. Were all randomized participants analyzed in the group to which they were allocated?	Yes	Unclear
Selective reporting	8. Were the groups similar at baseline regarding the most important prognostic indicators?	No	Yes
	9. Were cointerventions avoided or comparable?	Unclear	Unclear
	10. Was the compliance acceptable in all groups?	No	Yes
	11. Was the timing of the outcome assessment similar in all groups?	Yes	Yes
Total		4	6

Table 3B. Methodological quality of the included studies on safety

	First Author Year [Ref]								
Sources of Risk of Bias	Manchikanti 1999 [25]	Manchikanti 2012 [23]	Chun-jing 2012 [24]	Abejón 2007 [26]	Murai 2007 [27]	Avellanal 2008 [28]	Takeshima 2009 [29]	Pereira 2016 [30]	Lee 2014 [31]
1 Are the adverse events described?	Yes	No	Yes	No	Yes	Yes	No	Yes	No
2 Is the seriousness of the adverse events clearly stated?	Yes	No	Yes	No	No	No	No	Yes	No
3 A standardised method was used to detect and monitor adverse events?	?	?	?	?	?	?	?	?	?
4 Was the follow-up period adequate?	Yes	Yes	Yes	No	Yes	?	?	No	?
5 Were measuring instruments validated or standardized?	?	?	?	Yes	Yes	?	?	?	?
6 Were adverse events registered by the same person who provided treatment?	?	?	?	No	Yes	?	?	?	?
7 Was it clearly explained how the effect was attributed to the intervention/exposition?	No	Yes	Yes	No	Yes	Yes	No	Yes	No
8 Were numerical data per intervention group provided?	NA	Yes	Yes	NA	NA	NA	NA	NA	Yes
9 Were any adverse events, or some patients, excluded from the analysis on adverse events?	?	?	?	?	No	No	?	No	?

? = unclear; NA = not applicable.

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Table 4A. Results on effectiveness in the included studies comparing adhesiolysis and epidural injections

First Author Year [Ref]	Outcome	Measure	Period		Percutaneous Adhesiolysis	Epidural Injection	*P* Value
Manchikanti 2012 [23]	Pain	NRS, Mean (SD)	Baseline		8.1 (0.8)	7.9 (0.8)	0.001 (group difference), 0.001 (group by time interaction)
			3 mo		3.4 (0.8)	4.9 (1.6)	
			6 mo		3.7 (1.1)	5.8 (1.5)	
			12 mo		4.0 (1.2)	6.1 (1.4)	
			18 mo		3.6 (1.2)	6.1 (1.4)	
			24 mo		3.6 (1.1)	6.2 (1.4)	
		Patients with significant relief of >50%,* No.	3 mo		90	35	NR
			6 mo		85	18	NR
			12 mo		73	12	NR
	Disability	ODI, mean (SD)	Baseline		31.2 (4.1)	28.6 (4.1)	0.001 (group difference), 0.001 (group by time interaction)
			3 mo		15.2 (4.1)	20.2 (6.6)	
			6 mo		15.2 (5.2)	22.3 (6.1)	
			12 mo		15.8 (5.6)	23.3 (5.8)	
			18 mo		14.6 (4.6)	23.3 (5.7)	
			24 mo		13.9 (5.1)	23.2 (6.7)	
		Patients with significant relief of >50%,* No.	3 mo		92	37	NR
			6 mo		88	25	NR
			12 mo		77	13	NR
	Employment	Total employed, No.	Baseline		5	9	NR
			12 mo		5	9	NR
			24 mo		5	9	NR
		Eligible for employment,† No.	Baseline		5	12	NR
			12 mo		5	12	NR
			24 mo		5	12	NR
	Daily opioids	Morphine equivalents, mean (SD)	Baseline		99.0 (117.1)	69.6 (106.8)	0.149 (group difference), 0.485 (group by time interaction)
			3 mo		76.6 (99.8)	68.0 (91.4)	
			6 mo		75.1 (97.5)	67.9 (91.4)	
			12 mo		78.2 (103.7)	67.9 (91.4)	
			18 mo		76.3 (96.5)	67.9 (91.4)	
			24 mo		76.3 (96.5)	67.9 (91.4)	
Chun-jing 2012 [24]	Pain	VAS, mean (SD)	Baseline		6.95(1.25)	7.03 (1.24)	0.783
			1 mo		3.55 (2.16)	6.0 (1.74)	<0.001
			6 mo		3.71 (1.96)	6.21 (1.64)	<0.001
	Patient satisfaction	MacNab,‡ No.	6 mo	Excellent	8	0	<0.05
				Good	11	2	<0.05
				Common	7	17	<0.05
				Bad	12	19	<0.05
		MacNab,‡ %	6 mo	Excellent and good rate	50.00	5.26	NR
	Daily opioids	Morphine equivalents,§ mean (SD)	Baseline		65.26 (25.76)	64.74 (23.91)	0.927
			1 mo		32.63 (22.50)	51.05 (21.15)	<0.001
			6 mo		31.84 (21.29)	51.58 (22.12)	<0.001
Lee 2014 [31]	Pain in the back	NRS, mean (SD)	Baseline		6.4 (1.6)	6.4 (1.8)	0.986
		% of patients with successful outcomes	2 wk		65.4	53.2	0.252
			6 mo		55.8	33.9	0.023
	Pain in the leg	NRS, mean (SD)	Baseline		7.0 (1.4)	7.0 (1.6)	0.955
		% of patients with successful outcomes	2 wk		65.4	64.5	1
			6 mo		53.8	30.6	0.014
	Disability	ODI, mean (SD)	Baseline		51.6 (17.1)	46.9 (19)	0.167
		% of patients with successful outcomes	2 wk		63.5	50.0	0.186
			6 mo		59.6	29.0	0.001

NR = not reported; NRS = numeric rating scale (range from best to worst = 0–10); ODI = Oswestry Disability Index (range from best to worst = 0 to 50; not the usual range = 0–100); VAS = visual analog scale (range from best to worst = 0–10).

*Data extracted from Manchikanti 2009. Data on pain and disability at 18 and 24 months were undisclosed. An ad hoc average pain score is provided instead, which appears to combine the score on the NRS and the ODI. No *P* values for its comparison across groups are provided. The authors did not provide separate data.

†No housewives, no disabled, no one >65 years of age.

‡Excellent: disappearance of pain and numbness symptoms, without any motor dysfunction; good: disappearance of the most primary symptoms, occasional pain; common: symptom improvement, pain, and inability to participate in daily work; bad: nonimprovement or aggravation of symptoms, inability to participate in daily work.

§Time frame of assessments is inconsistent in the original paper. The author did not clarify these inconsistencies despite being repeatedly asked to do so.

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Table 4B. Results on effectiveness in the included observational studies

First Author Year [Ref]	Outcome	Measure	Period	Group		*P* Value
Manchikanti 1999 [25]	Pain	Patients with significant relief of >50% – 1st procedure, %	<1 mo	EA	100	NS
				NEA	100	
			1 mo	EA	97	<0.05 (group difference),
				NEA	72	<0.05 (group EA by time)
			3 mo	EA	80	<0.05 (group difference),
				NEA	25	<0.05 (group EA by time)
			6 mo	EA	52	<0.05 (group difference)
				NEA	10	
			12 mo	EA	22	<0.05 (group difference)
				NEA	7	
			≥12 mo	EA	8	NS
				NEA	5	
		Patients with significant relief of >50% – 2nd procedure, %	<1 mo	EA	100	NS
				NEA	100	
			1 mo	EA	94	NS
				NEA	92	
			3 mo	EA	88	<0.05 (group difference)
				NEA	46	
			6 mo	EA	75	<0.05 (group difference)
				NEA	22	
			12 mo	EA	25	NS
				NEA	10	
			≥12 mo	EA	0	NS
				NEA	4	
Abejón 2007 [26]	Pain	Mean	Baseline	E	8	NA
			4 wk	E	4	<0.001
			2 mo	E	5	<0.001
			3 mo	E	5	<0.001
Avellanal 2008 [28]	Pain	VAS, mean (SD)	Baseline	EA	7.89 (0.19)	NA
			4 wk	EA	5.00 (0.43)	0.001
			2 mo	EA	5.73 (0.42)	0.001
			3 mo	EA	5.95 (0.44)	0.001
			6 mo	EA	6.05 (0.46)	0.001
		Significant improvement (≥3-point reduction in the VAS), % (mean)	3 mo	EA	6 (3.7)	NR
			6 mo	EA	6 (3.3)	NR
		Mild to moderate improvement (1–2-point reduction in the VAS), % (mean)	3 mo	EA	7 (6.6)	NR
			6 mo	EA	7 (6.7)	NR
		Any improvement, % (mean)	3 mo	EA	6 (7.5)	NR
			6mo	EA	6 (8.0)	NR
Takeshima 2009 [29]	Disability	RDQ, mean (SD)	Baseline	E	13.6 (3.1)	NA
				E+R	12.1 (3.2)	NA
				R	12.2 (4.5)	NA
			4 wk after operation	E	4.3 (2.9)	0.0051
				E+R	7.2 (4.4)	0.0117
				R	5.6 (3.9)	0.0077
			12 wk after operation	E	8.8 (5.2)	0.0244
				E+R	7.6 (4.2)	0.0117
				R	7.1 (5.5)	0.0117
			24 wk after operation	E	11.0 (5.6)	0.2076
				E+R	8.9 (4.9)	0.1614
				R	7.0 (4.4)	0.0117
		ODI, mean (SD)	Baseline	E	23.5 (6.5)	NA
				E+R	24.0 (3.9)	NA
				R	21.9 (7.7)	NA
			4 wk after operation	E	11.9 (6.2)	0.0051
				E+R	13.6 (9.6)	0.0077
				R	13.9 (7.6)	0.0077

(continued)

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Table 4B. continued

First Author Year [Ref]	Outcome	Measure	Period	Group		*P* Value
			12 wk after operation	E	18.8 (5.3)	0.1551
				E+R	14.2 (5.9)	0.0077
				R	17.4 (7.2)	0.0180
			24 wk after operation	E	21.0 (5.5)	0.6241
				E+R	16.9 (5.1)	0.0209
				R	17.2 (7.4)	0.0117
		JOA, mean (SD)	Baseline	E	13.9 (4.4)	NA
				E+R	10.7 (1.8)	NA
				R	13.0 (6.6)	NA
			4 wk after operation	E	21.7 (3.7)	0.0051
				E+R	20.1 (5.9)	0.0077
				R	18.1 (6.3)	0.0077
			12 wk after operation	E	17.7 (3.4)	0.0663
				E+R	18.9 (4.5)	0.0077
				R	17.2 (5.7)	0.0117
			24 wk after operation	E	16.0 (4.8)	0.2626
				E+R	17.0 (4.7)	0.0077
				R	15.7 (6.3)	0.0173
Pereira 2016 [30]	Pain in low back	VAS, MD (95% CI)	1 d	EA	4.42 (2.44 to 6.41)	<0.001
			2 wk	EA	2.78 (0.81 to 4.75)	0.003
			4 wk	EA	3.53 (1.66 to 5.40)	<0.001
			3 mo	EA	2.71 (0.62 to 4.81)	0.007
			6 mo	EA	2.38 (0.27 to 4.49)	0.021
			12 mo	EA	2.21 (0.07 to 4.35)	0.040
	Pain in lower leg	VAS, MD (95% CI)	1 d	EA	4.44 (2.56 to 6.31)	<0.001
			2 wk	EA	3.75 (1.78 to 5.71)	<0.001
			4 wk	EA	3.04 (1.32 to 4.77)	<0.001
			3 mo	EA	3.01 (1.23 to 4.79)	<0.001
			6 mo	EA	3.39 (1.51 to 5.26)	<0.001
			12 mo	EA	2.27 (0.38 to 4.16)	0.013
	Disability	ODI, MD (95% CI)	2 wk	EA	16.84 (7.43 to 26.25)	<0.001
			4 wk	EA	11.28 (4.49 to 18.06)	<0.001
			3 mo	EA	8.02 (1.02 to 15.03)	0.019
			6 mo	EA	7.39 (–0.06 to 14.83)	0.053
			12 mo	EA	6.74 (–0.85 to 14.34)	0.101

E = epidural space; EA = endoscopic adhesiolysis; E+R = epidural space + nerve root; JOA = Japanese Orthopaedic Association Assessment of Treatment score; MD = mean difference; NA = not applicable; NEA = nonendoscopic adhesiolysis; NR = not reported; NS = not significant; ODI = Oswestry Disability Index 2.0; R = nerve root; RDQ = Roland-Morris Disability Questionnaire score; VAS = visual analog scale. CI = confidence interval

remains very low, even when data from the observational studies are taken into account.

Additionally, the cost-effectiveness of adhesiolysis is unknown, which implies that there is no evidence on whether covering the procedure in routine practice would be economically wise or sustainable. Generalization is further hampered by a lack of data on training standards and learning curve.

Furthermore, clinically significant adverse events have been reported after adhesiolysis, such as infection or bleeding [22,23,25,27,28,30]. This reporting derives from studies that have not clearly described how data on adverse events were gathered and that comprised a total sample of 240 patients who had undergone adhesiolysis. Therefore, it is impossible to reliably estimate the prevalence of the adverse events that have been reported and to rule out that other adverse events, which might remain unreported, exist. Further studies should systematically assess the type, characteristics, frequency, and severity of adverse events after adhesiolysis. Nevertheless, those reported so far are more serious than those from epidural injections.

The high number of reviews on adhesiolysis contrasts with the low number of published trials, which have been led by a small number of authors [11–16,37,38]. According to the two trials on effectiveness of adhesiolysis for FBSS, this procedure would be more effective than epidural injections. However, both the quality of these studies and the quality of their reporting are low. Moreover, the existing evidence does not allow the epidural injections to be considered a gold standard or appropriate comparator [39,40]. In fact, comparing the effectiveness of adhesiolysis with a procedure that has not shown superiority over placebo may be futile [41].

Conclusions from this systematic review contradict those of some previous reviews [11–16,37,38]. This fact may be explained by differences in scope and methodology; the current review focused specifically on the use of

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Table 5. Results on safety

	First Author Year [Ref]												
	Chun-jing 2012 [24]		Manchikanti 2012 [23]		Manchikanti 1999 [25]		Abejón 2007 [26]	Murai 2007 [27]	Avellanal 2008 [28]	Takeshima 2009 [29]	Pereira 2016 [30]	Lee 2014 [31]	
Group	I	GC	I	GC	G1	G2	I	I	I	I	I	GD	GF
No.	46	46	60	60	50	50	15	37	19	28	24	38	14
Event													
During the procedure													
Subarachnoid entries			7										
Postoperative weakness			2										
Rash and itching					3	3							
Weight gain in excess of 5 lb					1								
Subarachnoid puncture					4	7							
Subarachnoid blockade					2	4							
Suspected infection					0	8							
Confirmed infections					0	0							
Headache/neck pain								10			1		
Leg pain/sciatica								7					
Dural puncture								2	4				
Lumbago								1					
Apnea								1					
Bleeding											1		
Sensory deficit											1		
Transient neurological symptoms (headache and hypoacusia)									4				
After the procedure													
Wound pain								1					
Headache								2	1				
Total	0	0	9	0	10	22	0	24	9	0	3	0	0

G1 = percutaneous adhesiolysis group; G2 = endoscopic adhesiolysis group; GC = comparator group; GD = decompression group; GF = fusion group; I = intervention group.

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adhesiolysis for treating FBSS and used standard methods to gather and assess the available evidence [17]. Moreover, this review was undertaken by a multidisciplinary group of methodologists, researchers, and clinicians specializing in low back pain, among which only one actually performed the procedure, and whose income does not vary depending on the number of procedures performed or the generalization of the technology. This differs from previous primary studies and reviews that were published by the same group [22,23,25,33,34,37,38], four of which were published in a journal founded by the senior author [22,25,33,37].

Uncertainty on safety issues, poor quality of evidence on short-term effectiveness, absence of data on efficacy, efficiency, and long-term effectiveness, and potential concerns about transparency make the use of adhesiolysis for FBSS clinically and ethically controversial. In this scenario, the best interest of the patients and payers would be better served by thoroughly assessing the effectiveness of adhesiolysis for treating FBSS and, should it prove effective, by rigorously implementing it in routine practice [39]. This approach would imply conducting high-quality studies to assess the safety and efficacy (vs a sham procedure), as well as the effectiveness and cost-effectiveness, of adhesiolysis (vs treatments proven effective or comparing whether its addition to standard treatment improves outcomes) [40]. Should these studies lead to positive results, the procedure should then be used in routine practice complying with the indication criteria and training implemented in these studies and subject to postmarketing surveillance. Such surveillance should systematically gather data on safety and identify factors predicting clinical response, making it possible to refine indication criteria.

It might be argued that the nature of adhesiolysis makes it impossible to blind physicians to treatment allocation, which hampers following this assessment and implementation scheme. It might also be argued that any double-blind study on adhesiolysis might face a high number of dropouts in the middle term, as patients may abandon the trial to other supposedly effective interventions if pain persists [41]. Moreover, ethical considerations can arise when conducting studies with sham or nonactive treatments [42]. All of the above could encourage the use of pragmatic trials in some cases [41] or nudge clinicians into accepting the existing evidence, despite its limitations, for carefully selected patients when there are no other options. However, these considerations should not be an impediment for conducting new high-quality studies vs sham. It is indeed possible to blind patients and outcome assessors to treatment allocation, which makes it possible to conduct double-blind RCTs, and in fact, other nonpharmacologic interventionist treatments for low back pain have successfully followed this assessment and implementation scheme [42–45], which shows that it is feasible.

It might also be argued that such a scheme is unrealistic for procedures that are already used in routine practice and involve economic interests; in fact, many treatments are used in routine practice despite lacking any evidence on effectiveness, or even against evidence showing that they are ineffective [39,46]. Imposing a moratorium on coverage for procedures lacking evidence on effectiveness has been suggested as a way to overcome this problem [47].

The main limitation of this review is the low number and quality of available studies identified on the use of adhesiolysis for FBSS. However, this review has made it possible to highlight these facts, and the research agenda suggested above may overcome these limitations. This review may not include potentially relevant studies published in nonindexed journals that were not cited by those that were identified. However, nonindexed journals are less likely to publish high-quality studies, which might change the conclusion of this review. This review systematically gathered all the available evidence on adhesiolysis for FBSS, used recommended methods [17–19], and did not apply any date or language restrictions.

In conclusion, this systematic review shows that there is currently no evidence on the efficacy and cost-effectiveness of adhesiolysis for treating patients with FBSS and that the available evidence on effectiveness and safety is insufficient. Therefore, before generalizing adhesiolysis for this indication, high-quality RCTs should assess its efficacy, effectiveness, and cost-effectiveness, indication criteria and training standards should be established, and postimplementation surveillance mechanisms should be implemented to systematically gather data on safety and results in clinical practice.

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